



CLEF 2026 Working Notes (Jena, Germany)

CYUT at CLEF 2026 SimpleText Track: Hallucination Detection based on Retrieval-Aligned Contextual Fusion and Multi-Tier Cascaded Discernment

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Clinical Challenges & Objectives

Medical text simplification frequently suffers from **factual overgeneration** and ungrounded semantic **hallucinations**. We propose a defense framework coupling retrieval alignment and multi-tier cascaded discernment.



Three-Tier Core Defense Architecture

1. Fine-tuned **DeBERTa-v3** for retrieval-aligned semantic discernment & adaptive early exit.
2. **Qwen3-8B** computing Length-Normalized Contrastive Perplexity (LN-CP) for active auditing.
3. **XGBoost Ensembles** for clinical knowledge calibration & numerical symbolic guardrails.



Key Results & Ablation Findings

Official evaluation achieves Doc F1 = 0.8071 (Precision: 0.7980, Recall: 0.8163). Ablation confirms that removing the **retrieval module** causes Doc F1 to collapse to 0.4602, demonstrating external factual alignment as the foundation.

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Introduction & Motivation

Clinical value of medical text simplification, fatal hallucination risks in generative AI, and
CLEF 2026 SimpleText benchmark tasks

What is Medical Text Simplification?

Transforming jargon-heavy, statistic-dense abstracts from the Cochrane Library and professional literature into patient-friendly narratives to promote public health literacy.

Inviolable Principle: Simplification must never compromise factual accuracy!

Three Fatal Failure Modes in Generative AI

Dosage & Numerical Distortion: Arbitrarily hallucinating routine regimens into "500 mg daily for two weeks", endangering medication safety.

Clinical Polarity Reversal: Altering "no statistically significant difference" into "significantly effective", misleading efficacy.

Ungrounded Extraneous Injection: Fabricating unsupported claims regarding unrelated disease prevention, generating pseudo-science.

CLEF 2026 SimpleText Track Scope & Task Definition



Task 1: Scientific Simplification

Focuses on paragraph- and document-level scientific simplification, evaluating LLM text restructuring under readability constraints.



Task 2: Controlled Creativity

Task 2.1: Binary Overgeneration Detection

Task 2.2: Fine-Grained Taxonomy Classification

Task 2.3: Grounded Simplification by Design



Task 3: Research Classification

Predicting discipline & research field labels of scientific papers to optimize automated indexing for digital libraries (OpenAIRE, MADOC).

Official Findings: Evaluation Crisis & Document-Level Veto

The LLM "Evaluation Crisis"

Official statistics across 70 international system submissions reveal:

29% of submissions contain spurious sentences in at least 25% of input sentences.

21% of submissions exhibit hallucination in over 50% of sentences.

Failure of Traditional Metrics: Token-overlap metrics (BLEU/SARI) and LLM-as-a-judge are deceived by fluent grammar and fail to penalize unauthorized fabrications!

Official Rule: Document-Level Veto

To enforce zero-tolerance clinical standards:

If **any single sentence** within a document is flagged as overgeneration, via the logical OR operator, **the entire document is classified as positive (unacceptable violation)!**

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Related Work

Bidirectional language encoders, post-processing factuality verification, RAG probing mechanics, uncertainty estimation, and decision tree symbolic calibration

Foundational Language Encoders (BERT / RoBERTa / DeBERTa)

BERT and RoBERTa pioneered bidirectional continuous semantic representation extraction. DeBERTa disentangles content and relative position vectors, and integrates an Enhanced Mask Decoder to capture subtle word order shifts in complex clinical sentences.

Post-Processing Factuality Validation

In high-stakes medical settings, weight-free external auditing tools are highly practical (e.g., FactScore, CoNLI). Hierarchical Semantic Piece (HSP) inspired our approach of decoupling neural semantics from symbolic rules.



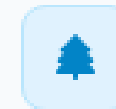
RAG Probing Mechanics (ReDeEP)

Literature confirms: when attention modules fail to lock onto retrieved facts, LLM parametric memory over-injects internal information to produce hallucinations. This motivates our strict Top-3 semantic anchoring before generation.



Uncertainty Estimation (LogTokU)

Standard Softmax normalization discards absolute evidence strength. LogTokU treats Logits as Dirichlet distributions; we expand this into Length-Normalized Contrastive Perplexity (LN-CP).



GBDT Symbolic Calibration (XGBoost)

Neural networks have blind spots in vector space. Using XGBoost decision forests to wrap rigid rules effectively smooths posterior probabilities, regularizing ambiguous predictions back into factual bounds.

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Approach

Retrieval-aligned contextual fusion, three-tier cascaded discernment pipeline, adversarial data augmentation, and multi-class taxonomy routing

Dataset Scale & Overgeneration Error Taxonomy

 Table 1: Dataset Partition Statistics

Data Split	Documents	Total Sentences	Positive Doc Ratio
Train	21,057	350,583	21.0% (4,427 docs)
Dev	1,120	18,638	20.9% (234 docs)
Test	6,728	105,001	*

 Table 2: Training Label Distribution & Overgeneration Taxonomy

Category Target	Count (Sentences)	Overgeneration Description
NONE	323,702	Safe sentence; no overgeneration
GENERATION_FAILURE	11,331	Catastrophic output, degenerate loops, truncation
LEAKED_INSTRUCTIONS	8,238	Leaked system prompt framing, thinking process
UNGROUNDED_INJECTION	5,735	Most Dangerous! Unsupported factual additions
REPETITIVE_CONTENT	1,477	Pathological repetition loops
GROUNDED_OVERGEN	100	Source-supported content beyond simplification scope



Adversarial Contextual Data Augmentation

➤ Adversarial Fishing Prompting

To prevent models from becoming blind to "fluent and professional-sounding fabrications", we used Qwen3-8B as an undercover generator, purposefully blending subtle fake dosages and prognosis assumptions to harvest 15,086 hard negative overgeneration sentences.

Fishing Core Directive: "Seamlessly blend in relevant clinical facts, modify numerical values and content to make it look professional and informative."

🔒 Strict Data Isolation & Filtering Protocol

Zero Token Overlap Guarantee: Strict exact-string matching verified zero overlap with Dev/Test splits.

Jaccard Divergence Check: Ensured synthetic vocabulary aligned closely with genuine medical distributions.

Dynamic Class Balancing: Because synthetic samples reached 3.4× baseline positives, dynamic sub-sampling was enforced to preserve gradient stability.

Three-Tier Cascaded Discernment System Architecture

Pre-Fusion

Contextual Fusion (RAG)

Dense retrieval extracts Top-3 most relevant facts, prepends source opening sentence (s_0) to assemble objective baseline evidence.

Layer 1

Semantic Discernment (DeBERTa)

Fine-tuned DeBERTa-v3-large cross-encoder. High-confidence safe samples trigger early exit, saving 80% compute.

Layer 2

Active Auditing (Qwen3)

For ambiguous grey-area sentences (0.35~0.75), Qwen3-8B computes Length-Normalized Contrastive Perplexity (LN-CP) to quantify information gain.

Layer 3

Knowledge Calibration (XGBoost)

XGBoost decision tree tracks 5D rigid rule matrix, inspecting numerical mismatch & negation polarity reversal for final calibration.

Why Does Isolated Sentence Verification Fail?

Clinical findings in biomedical abstracts span across multiple sentences. Evaluating a simplified candidate against a single source sentence out-of-context introduces extreme semantic passivation, causing false positives.

Advantages of Dual-Stream Alignment

- **Eliminating Semantic Blind Spots:** Allows models to see cross-sentence reasoning paths and multi-sentence deductions.
- **Unified System Baseline:** Ensures DeBERTa neural networks, Qwen causal models, and XGBoost ensembles operate on an identical objective factual ground-truth plane.

⚡ Layer 1: Semantic Discernment & Adaptive Early Exit

■ DeBERTa-v3-large Continued Fine-Tuning (CFT)

3:1 Dynamic Batch Balancing: Enforces a strict 3:1 majority-to-minority ratio during batch generation, preventing bias toward safe (NONE) labels.

Cosine LR Decay: Low learning rate ($\eta = 5 \times 10^{-6}$) with 10% linear warmup to stabilize dense medical representations.

Gradient Regularization: Gradient clipping optimization barrier ($\max \|\nabla L\| = 1.0$) prevents cross-attention instability.

■ Adaptive Early Exit Routing

Tri-state routing based on posterior probability P_{deb} :

Safe passage ($P_{deb} < 0.35$): Confirmed safe, output NONE to exit early, significantly reducing computational burden by 80%!

Violation interception ($P_{deb} > 0.75$): Confirmed over-generation, directly determine positive!

Gray blurry area ($0.35 \leq P_{deb} \leq 0.75$): Initiate Layer 2 deep auditing.

Layer 2: Active Generative Auditing & LN-CP Metric

Length-Normalized Contrastive Perplexity (LN-CP)

Deployed unquantized Qwen3-8B (Float16) to measure information gain:

$$\text{LN-CP} = (\text{Loss_no_ctx} - \text{Loss_ctx}) / (\text{Loss_no_ctx} + \epsilon)$$

Relative Information Gain Ratio: Measures the percentage of cross-entropy reduction directly attributable to external facts.

Suppressing Length Drift: Normalizing by Loss_no_ctx eliminates sentence length inflation, establishing length-invariance!

Computational Utility & Efficiency

Significant Information Gain: If perplexity drops sharply upon providing context, the sentence is source-supported; if unchanged, it is pure fabrication.

High Compute Savings: Only ~15%–20% of ambiguous sentences trigger the 8B model, balancing efficiency with extreme precision.

Layer 3: Clinical Knowledge Calibration & Symbolic Shield

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1. Numerical & Dosage Barrier

Compares numerical entities between simplified sentence and context. If unseen dosages, percentages, or numbers appear, triggers 1.0 rigid penalty!

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2. Logic Negation Polarity Reversal

Tracks clinical negation keywords (not, no, unable, unproven). If source negation disappears in simplified sentence, immediately flags polarity reversal!

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3. XGBoost Decision Arbitration

Regularized tree ensembles unify neural probabilities with rigid symbolic matrix, smoothing decision boundaries to ensure zero numerical omissions.

Downstream Multi-Class Categorization Strategies (Task 2.2)

Strategy 1: Weighted Cross-Encoder (ms-marco-MiniLM)

Configured with a 6-class output head and a two-stage anti-imbalance scheme:

Prior Static Sub-sampling: 25% retention cap on NONE examples to prevent majority dominance.

Inverse-Density Loss Weighting: Custom Weighted-LossTrainer with extra 1.2× prior weight for UNGROUNDED_INJECTION.

Strategy 2: Feature-Engineered LightGBM vs. Zero-Shot Control

10D Feature LightGBM: Targets prompt leakage and format collapse with 31 leaves and balanced weights.

Zero-Shot Control: Un-tuned Qwen3-8B achieves only 5.7% accuracy due to strict formatting sensitivity, highlighting the necessity of cascaded binary filtering.

 Symbolic Feature Engineering Matrix Compendium (5D vs 10D vs 13D)

Track Name	Feature Dimensions	Target Pathology Class	Algorithmic Principle
5D Deployed Track(Online Real-Time)	Length ratio, lexical overlap, new word count, numerical mismatch barrier, negation polarity variance	Task 2.1 binary defense, clinical logic reversal	Direct literal comparison against fused context; triggers immediate penalty upon numerical or negation mismatch.
10D Extended Track(Multi-Class Sorting)	5D features + repetition ratio, instruction leakage density, malformed geometry ratio, punctuation edge density, unique token ratio	Task 2.2 error diagnosis, prompt leakage, decoding loops	Scans for prompt meta-framing ("As an AI assistant..."), degenerate token repetitions, and corrupted geometries.
13D Independent Track(Standalone Analysis)	10D features + Shannon entropy, decision margin gap, logit space variance	Quantifying neural uncertainty and confidence dispersion	Measures prediction probability dispersion; triggers tighter validation boundaries when model exhibits hesitation.

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Experiments & Official Analysis

Task 2.1 binary detection leaderboard, ablation analysis, Task 2.2 multi-class categorization, and cross-team benchmark comparisons

🏆 Task 2.1 Binary Hallucination Detection Official Benchmark

Architecture Configuration / Baseline	Doc Prec.	Doc Rec.	Doc F1	Sent Prec.	Sent Rec.	Sent F1
CYUT Complete (DeBERTa_XGB_QwenAudit)	0.7980	0.8163	0.8071	0.8367	0.8855	0.8604
Ablation: Retrieval + DeBERTa + XGB (Without QwenAudit)	0.7820	0.8200	0.8005	0.8274	0.8855	0.8555
Ablation: Without Retrieval Component	0.3045	0.9421	0.4602 (<i>Collapse</i>)	0.1977	0.9072	0.3246
Ablation: Without XGBoost Classifier	0.6934	0.8889	0.7791	0.7567	0.9279	0.8336
Baseline: Retrieval + BERT Base	0.6176	0.9017	0.7331	0.7035	0.9353	0.8030
Baseline: Retrieval + RoBERTa Base	0.7135	0.8714	0.7846	0.8397	0.7713	0.9216
Baseline: Retrieval + DeBERTa Base	0.6380	0.8044	0.7116	0.7022	0.8340	0.7116
Ensemble: Triple Ensemble	0.8072	0.8035	0.8053	0.8379	0.8838	0.8603

Core Finding: Retrieval Alignment contributes over **43% of system performance!** Removing Top-3 contextual fusion deprives models of factual anchors, reducing predictions to random guesswork.

 Task 2.2 Fine-Grained Multi-Class Categorization Results

Cascaded Configuration / Downstream Sorter	Accuracy	Doc Prec.	Doc Rec.	Doc F1	Sent F1
Proposed Synergy: Task 2.1 (Optimal-Recall) + Cross-Encoder	0.8037	0.7425	0.8421	0.7892	0.8441
Task 2.1 (Config-C) + Cross-Encoder Tuning	0.7859	0.7980	0.8163	0.8071	0.7859
Task 2.1 (Config-B) + Cross-Encoder Tuning	0.7792	0.7952	0.7952	0.7952	0.8556
Task 2.1 (Config-A) + LightGBM Tree Ensemble	0.7096	0.7820	0.8200	0.8005	0.8555
Task 2.1 (Config-A) + Qwen3-8B (Zero-Shot)	<u>0.0570</u>	0.7820	0.8200	0.8005	0.8555
Only ms-marco-MiniLM-L-6-v2	0.7878	0.5588	0.8421	0.6718	0.7502

***Zero-Shot Paradox Interpretation:** Qwen3 zero-shot accuracy collapses to 5.7% due to strict formatting sensitivity, yet inherits the high-quality upstream binary filter (Config-A) to retain Doc F1 at 0.8005!

 Cross-Team Benchmarking (CLEF 2026 Overview)

Team & Core Methodology	Task 2.1 Doc F1	Task 2.2 Accuracy	Leaked Instr. Recall	Gen. Failure Recall	Ungrounded Inj. Recall
CYUT (DeBERTa + Qwen + XGB)	0.8071	80.37%	96%	97%	74%
AIIRLab (DeBERTa + LLaMA3.1 Majority Vote)	0.8200	85.00%	99%	98%	74%
Writerslogic (DeBERTa + LightGBM Stack)	0.8100	81.00%	96%	96%	73%
DUTH (Biomedical BERT Ensembles)	0.8000	79.00%	98%	97%	69%
FutureMakers (TF-IDF + PubMedBERT)	0.7800	79.00%	95%	97%	67%

***Critical Performance Watershed:** All top teams recall Prompt Leakage and Generation Failure at >95%; the decisive factor lies in capturing **Ungrounded Injections (UNGROUNDED_INJECTION)**!

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Discussion & Conclusions

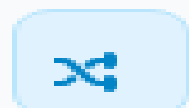


OUR experiences



High-Difficulty Error Breakdown

Ungrounded Injection: Highly fluent grammar with no obvious flaws. We leveraged adversarial fishing training to elevate detection recall to the top-tier 74% benchmark.



Rare Pathology Bottlenecks

Grounded Overgeneration: With only 100 training examples (0.03%), all participating teams achieved recall below 40% due to extreme sample scarcity.



Architectural Practice Insights

Tri-Tier Efficiency: Lightweight neural layer filters 80% normal texts, LLMs handle ambiguous grey zones, and decision trees safeguard rigid numerical rules.

"Retrieval & Source Alignment" is the Lifeline of Hallucination Governance

Systems exemplified by ours and FutureMakers (paper462) empirically demonstrate that a RAG mechanism which segments long documents into individual **sentences and maintains strict contextual alignment across both training and inference** fundamentally resolves the rigid bottleneck of **token truncation inherent** in neural models. Our ablation study, which shows an **F1-score plummeting to 0.4602**, provides the most definitive empirical evidence for this conclusion.

Shifting LLMs from Naive Judges to **Deep Semantic Entropy Profilers**

Previous systems merely employs an LLM as a naive label classification judge (**LLM-as-a-judge**), leaving it highly susceptible to **prompt sensitivities and hallucination biases**. In contrast, our **Length-Normalized Contrastive Perplexity (LN-CP)** metric treats the LLM as a **grey-box information entropy extractor** to quantify the **relative information gain ratio**, thereby offering a significantly **more objective and robust semantic measurement**.



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Thanks for your attention

